COMPLETION MAPS FROM ETALE-OVER-COMPLEMENT ALGEBRAS COME FROM THE HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: uniqueness, the factorization assertion and existence via the reduction A = C

Editorial note. The label lemma-fully-faithful-etale-over-complement names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- I-adic completions and the henselization of a Noetherian pair as a filtered colimit of etale algebras preserving the reduction
- the cotangent complex exact sequence for compositions and the annihilation of its cohomology in the rig-etale case
- idempotent-quotient lemma identifying a surjective completion with the completion of a finite type algebra
- etale-ness checked on infinitesimal neighbourhoods
- the universal property of the henselization

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 2103; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ALQ: sources/stacks-restricted/pinned-original-source.tex, source lines 1981–1981. Prerequisite pointer invoked at proof line(s) 2103; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A02: sources/stacks-chow/tags.txt, source lines 12620–12620. Prerequisite pointer invoked at proof line(s) 2076; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGV: sources/stacks-chow/tags.txt, source lines 13208–13208. Prerequisite pointer invoked at proof line(s) 2084; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0ALR: sources/stacks-chow/tags.txt, source lines 13379–13379. Prerequisite pointer invoked at proof line(s) 2110; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A43: sources/stacks-chow/tags.txt, source lines 12761–12761. Prerequisite pointer invoked at proof line(s) 2119; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let B be a finite type A-algebra such that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is étale over $\mathrm{Spec}(A) \setminus V(I)$. Let C be a Noetherian A-algebra. Then any A-algebra map $B^\wedge \to C^\wedge$ of I-adic completions comes from a unique A-algebra map*

$$B \longrightarrow C^h$$

where C^h is the henselization of the pair (C, IC) as in More on Algebra, Lemma 0A02. Moreover, any A-algebra homomorphism $B \to C^h$ factors through some étale C-algebra C' such that $C/IC \to C'/IC'$ is an isomorphism.

Proof. Uniqueness follows from the fact that C^h is a subring of $C^\wedge$, see for example More on Algebra, Lemma 0AGV. The final assertion follows from the fact that C^h is the filtered colimit of these C-algebras C', see proof of More on Algebra, Lemma 0A02. Having said this we now turn to the proof of existence.

Let $\varphi : B^\wedge \to C^\wedge$ be the given map. This defines a section

$$\sigma : (B \otimes_A C)^\wedge \longrightarrow C^\wedge$$

of the completion of the map $C \to B \otimes_A C$. We may replace (A, I, B, C, φ) by $(C, IC, B \otimes_A C, C, \sigma)$. In this way we see that we may assume that $A = C$.

Proof of existence in the case $A = C$. In this case the map $\varphi : B^\wedge \to A^\wedge$ is necessarily surjective. By Lemmas 0ALQ and 00S2 we see that the cohomology groups of $NL^\wedge_{A^\wedge/_\varphi B^\wedge}$ are annihilated by a power of I. Since φ is surjective, this implies that $\mathrm{Ker}(\varphi)/\mathrm{Ker}(\varphi)^2$ is annihilated by a power of I. Hence $\varphi : B^\wedge \to A^\wedge$ is the completion of a finite type B-algebra $B \to D$, see More on Algebra, Lemma 0ALR. Hence $A \to D$ is a finite type algebra map which induces an isomorphism $A^\wedge \to D^\wedge$. By Lemma 0ALQ we may replace D by a localization and assume that $A \to D$ is étale away from $V(I)$. Since $A^\wedge \to D^\wedge$ is an isomorphism, we see that $\mathrm{Spec}(D) \to \mathrm{Spec}(A)$ is also étale in a neighbourhood of $V(ID)$ (for example by More on Morphisms, Lemma 0A43). Thus $\mathrm{Spec}(D) \to \mathrm{Spec}(A)$ is étale. Therefore D maps to A^h and the lemma is proved. □